Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which of the following feature attribution methods provides pixel-level importance? 1 / 1 point

- Grad-CAM
- Vanilla Gradient
- Guided Grad-CAM

Correct
Correct! Vanilla Gradient provides pixel-level importance.

2. Which of the following are cons of using feature visualization (select all that apply)? 1 / 1 point

- [x] For large networks, it is difficult to visualize the complete network

Correct
Correct! A con of feature visualizations is that, for large networks, it is difficult to visualize the complete network.

- [x] Many feature visualization images contain some features with no human interpretation

Correct
Correct! Many feature visualization images contain some features with no human interpretation, which is a con of using feature visualization.

- [x] It is not a complete picture of interactions

Correct
Correct! A con of feature visualizations is that they are not a complete picture of interactions in a neural network

3. Which explanation approach requires datasets that are labeled on the pixel level with the concepts? 1 / 1 point

- DeconvNet
- Network Dissection
- Concept Activation Vectors

Correct
Correct! A challenge with using network dissection is that you need datasets that are labeled on the pixel level with concepts.

4. Which explanation approach requires the curation of multiple datasets? 1 / 1 point

- DeconvNet
- Network Dissection
- Concept Activation Vectors

Correct
Correct! Concept Activation Vectors require the curation of multiple datasets, including a random dataset and concept dataset(s).

5. It is best practice to train multiple CAVs using different random datasets while keeping the concept dataset the same. How many N random datasets are recommended by the original study authors? 1 / 1 point

- N>=3
- N>=10
- N>=1000

Correct
Correct! The original study authors recommend that N be greater than or equal to 10 for robustness.

6. TCAV performs poorly on what type of neural network? 1 / 1 point

- Shallow neural networks
- Deep neural networks
- Neural networks with residual connections

Correct
Correct! TCAV performs poorly on shallow neural networks because concepts in deeper layers are more separable.

7. Which of the followings are reasons why attention in explainability/interpretability is hotly debated (check all that apply)?: 1 / 1 point

- [x] Attention is often uncorrelated with gradient-based feature importance measures

Correct
Correct! A 2019 paper demonstrated that attention is often uncorrelated with gradient-based feature importance measures

- [x] A completely different set of attention weights that results in the same prediction can usually be found

Correct
Correct! A 2019 paper showed that a completely different set of attention weights that results in the same prediction can usually be found

- [x] One cannot intervene on attention while keeping all the other variables invariant

Correct
Correct! A 2020 paper demonstrated that one cannot intervene on attention while keeping all the other variables invariant

8. In "The elephant in the interpretability room" paper, what saliency methods were proposed as alternatives to explainable attention (check all that apply)? 1 / 1 point

- [ ] LIME
- [x] Layer-wise Relevance Propagation (LRP)

Correct
Correct! Propagation-based methods like LRP were proposed by the paper as an alternative to explainable attention methods.

- [x] Gradient-based methods

Correct
Correct! Gradient-based methods like Integrated Gradients were proposed by the paper as an alternative to explainable attention methods.

9. What explainable method is defined as the path integral of the gradients along the straightline path from the baseline x' to the input x? 1 / 1 point

- Integrated Gradients
- Layer-wise Relevance Propagation (LRP)
- Occlusion-based Methods

Correct
Correct! Integrated Gradients are defined as the path integral of the gradients along the straightline path from the baseline x' to the input x.

10. Bipartite Graph Attention Representation is an example of: 1 / 1 point

- Visualizing Attention Patterns
- Visualizing Occlusion-based Methods
- Visualizing Gradient-based Methods

Correct
Correct! A method proposed in 2018 for visualizing attention patterns is the Bipartite Graph Attention Representation.